

A review of the performance and structural considerations of paraffin wax hybrid rocket fuels with additives



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ARTICLE INFO

Keywords:

Hybrid rocket
Regression rate
Paraffin wax
Additives

ABSTRACT

Paraffin wax as a hybrid rocket fuel has not been comprehensively characterised, especially regarding the structural feasibility of the material in launch applications. Preliminary structural testing has shown paraffin wax to be a brittle, low strength material, and at risk of failure under launch loading conditions. Structural enhancing additives have been identified, but their effect on motor performance has not always been considered, nor has any standard method of testing been identified between research institutes. A review of existing regression rate measurement techniques on paraffin wax based fuels and the results obtained with various additives are collated and discussed in this paper. The review includes 2D slab motors that enable visualisation of liquefying fuel droplet entrainment and the effect of an increased viscosity on the droplet entrainment mechanism, which can occur with the addition of structural enhancing polymers. An increased viscosity has been shown to reduce the regression rate of liquefying fuels. Viscosity increasing additives that have been tested include EVA and LDPE. Both these additives increase the structural properties of paraffin wax, where the elongation and UTS are improved. Other additives, such as metal hydrides, aluminium and boron generally offer improvements on the regression rate. However, very little consideration has been given to the structural effects these additives have on the wax grain. A 40% aluminised grain, for example, offers a slight increase in the UTS but reduces the elongation of paraffin wax. Geometrically accurate lab-scale motors have also been used to determine the regression rate properties of various additives in paraffin wax. A concise review of all available regression rate testing techniques and results on paraffin wax based hybrid propellants, as well as existing structural testing data, is presented in this paper.

1. Introduction

Hybrid rocket technology has advanced since the discovery of liquefying fuels, also referred to as non-classical fuels. Hybrid rockets are named as such due to their propellants, which often comprise a solid fuel and a liquid or gaseous oxidiser. The physical and state separation of the fuel and oxidiser offer an increased safety aspect to rocket design and fuel handling. However, there is compromised performance due to the limiting boundary layer combustion method. The measure of regression rate of the solid fuel is a good indicator of the combustion performance of a propellant combination, which defines how quickly the solid fuel burns, however, the density of the propellant should also be considered. Hybrid fuels, particularly traditional variants such as hydroxyl-terminated polybutadiene (HTPB) are known to have very low regression rates. The regression rate is often improved through the use of performance-enhancing additives, such as high-energy metal particles. The degree to which these additives improve the combustion performance and the

effect they have on the structural performance on a solid fuel grain has not been comprehensively determined and is dependent on factors such as particle size and density, as well as the original binder material. Although we can consider the propellant performance independently, the performance of the rocket itself is dependent on various design parameters, such as vehicle size, design thrust, mass and fuel density. In this paper, we review the current state of paraffin wax fueled hybrid rocket motor performance testing with specific reference to the role of additives in improving metrics such as specific impulse and density impulse. The purpose of this review is to create a link between existing regression testing results, with various additives, in comparison to the available structural characterization data on those fuel mixtures of paraffin wax based hybrid fuels to determine their feasibility in launch applications.

2. Hybrid combustion models

Combustion within a hybrid rocket motor (HRM) differs from that

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Nomenclature

a	regression rate coefficient
A_b	combustion area (m^2)
G_o	oxidiser mass flux ($\text{kg}/\text{m}^2\text{s}$)
ID	inner diameter
I_{sp}	specific impulse
m	mass (kg)
n	flux exponent
OD	outer diameter
O/F	oxidiser to fuel ratio
$\dot{r}$	regression rate (mm/s)
r_f	regression rate (mm/s)
t_b	combustion time (s)
ρ_f	fuel density (kg/m^3)
HTPB	hydroxyl-terminated polybutadiene
HRM	hybrid rocket motor
SRM	solid rocket motor
LRM	liquid rocket motor
PMMA	polymethyl methacrylate
GOX	gaseous oxygen

within a solid rocket motor (SRM) with the separation of the oxidiser from the solid fuel grain, which resides in the combustion chamber. Combustion initiates when the oxidiser is either pumped or allowed to flow under pressure past the fuel grain surface, through either a single port or multiple ports. The number of ports is determined by the regression rate of the fuel material, and the required thrust. This combustion mechanism allows for the development of a flame zone above the exposed fuel surface and varies along the length of the port due to the changing O/F ratio. Classical fuels such as HTPB have very low regression rates as a result of the dominant boundary layer combustion mechanism. Non-classical fuels, such as paraffin wax have a mechanical combustion mechanism referred to as droplet entrainment in addition to the boundary layer mechanism, which promotes faster regression of the fuel. The focus of this review is the effect of additives on the non-classical combustion mechanism of paraffin wax through empirical means. However, a brief overview of both mechanisms is included for reference.

2.1. Classical hybrid combustion

Boundary layer combustion is a result of fuel vaporisation. The primary combustion region is shown to be within a relatively narrow flame zone located within the boundary layer [1], depicted in Fig. 1. The various forms of heat transfer allowing for fuel vaporisation include convection and radiation from the flame zone. Vaporised fuel is transferred away from the heated fuel surface towards the flame zone. Un-combusted oxidiser is moved from the main oxidiser stream to the

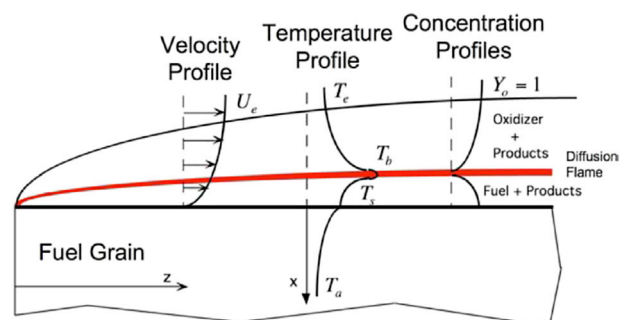


Fig. 1. Schematic of the classical hybrid fuel combustion mechanism [5].

flame by turbulent diffusion. The stoichiometric conditions of the reaction define the position of the flame within the boundary layer. The rate of the combustion reaction, which is dependent on pressure and temperature, determines the flame thickness. Other factors which affect the development of the boundary layer include chamber pressure, gas temperature, gas composition, mass flow rate, port length, and port diameter [2]. Classical combustion is limited by the diffusive heat transfer to the fuel surface, resulting in slow regression rates. There has been significant work conducted in the diffusion limited classical combustion mechanism defining the regression rate of these types of fuels and the reliance on the position of the flame zone within the boundary layer as well as the heat of gasification. This problem is highly complex and is beyond the scope of this review, however detailed work on the heat and mass transfer between chemically reacting liquids has been conducted by numerous researchers leading to adequate regression rate laws [3,4].

There have been some methods discussed for testing and enhancing the regression rate of classical hybrid fuels. To name a few, authors such as Gariana et al. developed a 2D HTPB/GOX slab motor to compare combustion results to a numerical simulation developed [6]. Li et al. investigated the increase in the average regression rate when using swirl injectors. The impinging injectors and oxidiser swirl resulted in an uneven burn [7]. Pei et al. considered the effect of altering the combustion chamber port geometry on the regression rate and combustion efficiency of the motor [8]. These modifying techniques indicated an increase in performance measures, but the effects did not result in a performance enhancement similar to that of high regression rate fuels which do not rely primarily of diffusive heat transfer.

2.2. Non-classical hybrid combustion

The identification of higher regression rate fuels has allowed hybrid rocket motor technology to develop further, addressing more practical requirements. The fundamental difference between the classical and non-classical combustion mechanisms is the formation of a low viscosity liquid layer on the regressing surface. These fuels, known as liquefying fuels, produce a thin, low viscosity, low surface tension liquid layer on the fuel surface during combustion. Instability of this liquid layer caused by the flow of oxidiser over the liquid surface results in the formation of waves, which promote the entrainment of droplets into the gas stream, increasing the overall mass transfer rate and combusting surface area. A schematic of this combustion mechanism can be seen in Fig. 2 [5].

The droplet entrainment significantly enhances the speed of regression, and thus addresses some of the classical hybrid performance concerns. The increased regression rate of these fuels reduces the need for multi-port designs, and thus overly complex geometries and manufacturing techniques. This fuel diffusion invention was patented by Karabeyoglu et al. [9,10] in which they developed a method of identifying hydrocarbon fuels that are solid at room temperature, have a mean carbon number between 15 and 80, and have a low molecular weight

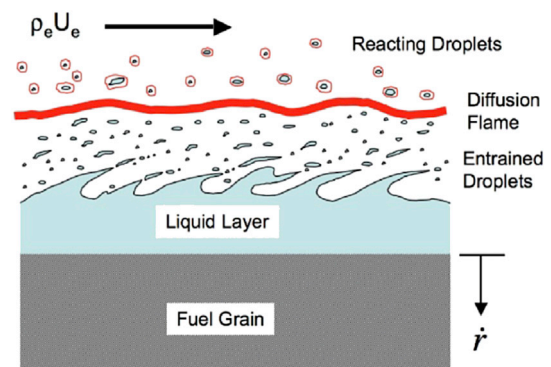


Fig. 2. Liquefying fuel entrainment mechanism [5].

[9,10]. An additional advantage of this regression enhancing combustion mechanism is that droplet entrainment is not affected by blowing as it is not a mechanism of the flame zone heat transfer [11]. Mazzetti et al. have conducted an extensive market review on the plausibility of hybrid rocket propulsion in the mid-term future, particularly since the discovery of liquefying fuels such as paraffin wax [12]. There has been extensive research conducted on the modelling of droplet formation and the combustion process of liquefying fuels considering non-equilibrium and quasi-equilibrium evaporation conditions. The equilibrium model can be considered for large droplet size. However, a non-equilibrium model is necessary for accuracy in the case of small droplets [13–15]. The development of the droplet formation and the size of the droplets formed for paraffin wax fuels will determine the increased regression rate of the fuel due to droplet entrainment. The focus of this review is the empirical solution to the regression rates of liquefying fuels, but test data should be validated against accurately determined liquid combustion models.

3. Performance enhancing additives

An advantage of making use of hybrid propellants with a solid fuel is the ability to include energetic additives in the casting process. Although liquefying fuels already show an improved regression rate, the use of energetic additives can further improve this, and potentially offer control over the regression rate [16]. While, metal additives, such as aluminium can increase nozzle erosion, certain metal hydrides, such as magnesium hydride, can reduce this effect, by reducing the oxidising species in the nozzle [16,17]. Additives which affect the material strength could be added to improve the structural performance of the fuel grain if it is prone to failure. However, this may alter the fuel liquid viscosity, and reduce the effect of entrainment regression. Consideration should be given to the effect on both the structural and performance properties of a fuel with the addition of any additive. This is apparent from studies where additives offered structural benefits, but in turn increased the melt layer viscosity, reducing entrainment and thus regression rate [18]. Classical hybrid fuels have little risk of structural failure due to their toughness and strength [19]. Newer fuels, such as paraffin wax, are comparatively brittle and weak and may be more likely to fail in use, necessitating a further investigation into their performance in the presence of thermal, pressure and dynamic loading [19].

3.1. Aluminium

Aluminium additives, both on a micro- and nano-scale have been shown to increase the regression rate of hybrid fuels while serving as a dense energy source [20–29]. The effectiveness of the additive creates an opportunity to make use of other oxidisers, such as nitrous oxide. Data has shown that the use of nitrous oxide with an aluminised HTPB grain can yield a similar regression rate to that of pure HTPB in gaseous oxygen (GOX) [30]. There are numerous advantages to using nitrous oxide (N₂O) such as self-pressurisation, low cost and availability. An added benefit of using aluminium powder is the reduced O/F ratio, resulting in a reduced oxidiser tank size [16].

The inclusion of aluminium particles within fuels can lead to an increase in specific impulse, volumetric heat of oxidation, adiabatic flame temperature, heat of combustion, and radiative heat transfer [31]. On a micro-scale, aluminium particles increase the regression rate by improving the radiative heat flux from the diffusion flame zone to the fuel surface as a result of the radiating metallic particulate matter and the higher gas-phase temperature [19]. Particles also improve the regression rate through the release of energy during metal oxidation [31]. In some studies, the addition of aluminium particles in the fuel grain does not significantly benefit the regression rate due to the high melting temperatures associated with the aluminium oxide layer, which coats the aluminium surfaces. The fuel often melts before the aluminium, creating a protective film around the particles, and further impeding the combustion process. The aluminium can be lost in the exhaust, and have little

benefit on the regression rate. This scenario is particularly prevalent in oxygen-rich environments, such as the use of a GOX oxidiser. The use of fluorine-based oxidisers reduced the build-up of the oxide layer and allowed for easier combustion [24].

Nano-sized aluminium particles are known to react differently to micro-sized particles. This can result in higher combustion efficiencies, better heat transfer, and a reduced ignition delay and burn time when compared to micro-sized particle-enriched grains [19]. The cost and manufacturing complexities involved are some of the disadvantages associated with the use of nano-sized particles. The reactive effect is similar to that of the micro-sized aluminium particles, however, with the lower ignition time and temperature due to the high specific surface area, the energy release is closer to the fuel surface [31]. Production is further complicated by the unreactive, oxide mass fraction which can develop around each particle if not processed in inert environments.

There is a limit to how much aluminium can be introduced to a propellant. The inclusion of aluminium particles increases the regression rate of that propellant combination, but when an excessive amount is added, the near-surface combustion becomes less efficient due to a build-up of particles at the fuel surface [19]. The mechanism behind the combustion of the aluminium particles themselves relies on their entrainment into the oxidiser stream. The particles can collect on the regressing fuel surface before they are forced into the oxidiser stream, resulting in a reduced regression rate due to limited heat transfer, especially in larger hybrid motors [31]. It is also noted that reducing the aluminium particle size increases regression rate insofar as the oxide layer formation is prevented. If methods of altering the oxide layer and coating the particles are used, the regression rate can be increased [31]. The inclusion of aluminium additives in hybrid fuels has been considered in applications such as the proposed hybrid design of the Europa and Uranus mission by Jens et al. and the Mars Ascent vehicle design by Chandler et al. The first design considers the addition of 30% micron-sized aluminium particles in PE wax [32], while the second considers the addition of 40% micron-sized aluminium particles to paraffin wax [33]. The intended advantage is such that the specific impulse and density impulse will increase, and the optimal O/F ratio will change. They note, however, that the theoretical performance may not be applicable due to the inefficiency attributed to aluminium combustion, reducing the overall combustion efficiency, and the slag formation in the post-combustion chamber.

3.2. Boron

Boron is considered to improve regression due to its higher volumetric heat release than aluminium, but its use is limited due to its difficulty with ignition which can lead to low combustion efficiencies [34]. Small amounts (5%) of boron additives in HTPB have shown a reduction in regression rates due to this ignition complexity [31]. Tests conducted at Pennsylvania State University (PSU) include the use of boron and boron carbide particles within HTPB fuels. When the additive amount in HTPB is near 10%, there is an increase in regression rate as compared to pure HTPB combustion [30]. Extensive boron testing has been conducted by the Israel Institute of Technology for use in solid fuel ramjets [35–37]. The difficulties in the combustion of boron and boron carbide is a known problem with the utilisation of this additive. An afterburner can be introduced to assist with full combustion. The boron particles can also be coated with materials such as magnesium, to assist with the combustion process.

3.3. Metal hydrides

Metal hydrides, such as magnesium hydride (MgH₂) and lithium aluminium hydride (LiAlH₄) have been considered for performance enhancement and their use results in an increase in the regression rate of paraffin wax [38]. Lithium-based particles are extremely reactive. However, the volumetric heat of oxidation is low compared to that of

aluminium or boron. Testing which includes LiAlH_4 as an additive in paraffin wax has shown a noticeable increase in regression rate [39]. Tests conducted by Smoot and Price [40] show that the amount of lithium hydride (LiH) added to an HTPB fuel affects the primary combustion mechanism and the pressure dependence of the combustion. At increasing oxidiser mass flow rates, the combustion mechanism moves from the pressure independent diffusion controlled combustion to the pressure dependent kinetic controlled combustion. In this case, higher levels of LiH within the fuel grain cause the transition range to extend, and the pressure sensitivity of the combustion to reduce.

3.4. Magnesium

Magnesium is a readily available and affordable metal which has similar thermal capabilities to that of LiAlH_4 , although it is slightly less reactive. It is easily ignited and burns with a hot flame, which can aid in the development of a liquid layer on the fuel surface [39]. Testing conducted by Schultz [39] revealed that the regression rate of paraffin wax could be improved with the addition of more than 10% MgH_2 . Amounts less than this result in a slight reduction in the regression rate as a consequence of the newly opaque grain surface reducing radiative heat transfer into the fuel.

3.5. Polymers

Polymers such as low-density polyethylene (LDPE) and ethylene-vinyl acetate (EVA) have been investigated with paraffin wax. In small amounts, they showed an improvement in the structural properties of the grains, but the regression rate was affected [18,41,42]. These polymers increase the melt viscosity of the wax, reducing the effect of droplet entrainment, and thus the regression rate. The increase in structural performance of grains with LDPE indicates a reduced risk of grain slumping due to storage [18].

4. Density impulse

Density impulse is a secondary measure of propellant performance. The specific impulse (I_{sp}) of a system is defined as the total impulse per unit weight, while the density impulse is defined as the total impulse per unit volume [43], and is often used as a performance measure in volume-strict designs [44,45]. Propellants with similar specific impulses can result in a different density impulse based on the propellant density. This results in a correlation between the size of the vehicle that is necessary to house the propellant. Energetic additives can be used to change this measure of performance, either resulting in a higher or lower density fuel. Generally, HRMs have a higher I_{sp} than SRMs, and a higher density impulse than liquid rocket motors (LRMs) [46]. The advantage of the density impulse may be negligible due to the grain sliver fraction, but this is dependent on the HRM port geometry [46].

Karabeyoglu et al. [44] considered an upper stage hybrid motor design to advance the technology readiness level (TRL) of advanced hybrid motors. In doing so, they derived a correlation between performance measures such as I_{sp} and density impulse to determine the performance comparison between HRMs, LRMs, and SRMs. With the addition of energetic particles to HRMs, there is a noticeable increase in I_{sp} , and density impulse, making them feasible for use when compared to LRMs [44].

Further investigation into performance enhancement of metal additives has led to a comparative study on the theoretical specific impulse at the optimal O/F ratio, and resultant density impulse for a 40% weight addition of various additives. The addition of metal additives (Al, B, Mg, Mg_2B_3) into a paraffin wax has been shown to slightly decrease the optimal specific impulse when using high-performance oxidisers, such as liquid oxygen (LOX) [47]. This is attributed to the increased dissociation effect at high temperatures, and the average molecular weight of the combustion products. There is, however, a noticeable increase in the

specific impulse for medium- and low-performance oxidisers (N_2O_4 and N_2O respectively) [47]. The density impulse for these metals, determined with the O/F ratio for the optimal specific impulse, increases for the high- and low-performance oxidisers, while decreases slightly for the medium-performance oxidiser. Metal hydrides (AlH_3 , MgH_2 , LiAlH_4 , LiAlH_4) indicate an increase in specific impulse for all the oxidisers. However, the Li-based additives tend to decrease the density impulse, most likely because of the element's low density [47].

5. Combustion testing

Often the regression rate of a propellant is used as a measure of propellant performance. Although this is not the only performance metric, it offers a useful basis for comparison between fuels. The regression rate is dependent on design-specific properties as well as propellant performance properties. A simplified equation which can be used to estimate regression rate, $\dot{r}$, as a function of oxidiser mass flux, G_o , on the basis of empirically-derived ballistic coefficients, a and n :

$$\dot{r} = aG_o^n \quad (1)$$

Techniques for quantifying the regression rate of fuels include the instantaneous measurement of fuel depletion throughout the burn or the time-averaged regression rate which is estimated based on the burn time, fuel mass, and flow rate. Instantaneous regression is determined with the use of slab motor test stands that allow a small section of propellant to be combusted with visualisation by transparent windows or x-ray to measure the instantaneous fuel regression [9,29,30]. The fuel grain height is assessed continuously throughout the combustion process. This technique has several disadvantages, not least that the resultant ballistic coefficients are affected by the geometry of the fuel slab as well as the interaction of the oxidiser flow and the side walls, or the effect of side burning [41]. Inaccuracies may also arise from the short burn times due to ignition transients. The primary use of a slab motor, however, is to compare the regression behaviour of pure fuels and those with additives and to determine the mechanisms behind the combustion process visually. This is especially informative when considering the liquefying fuel entrainment mechanism, which has since been visually identified in numerous slab motor tests.

A more physically representative method of determining the regression rate of a propellant combination is through the use of small-scale or full-scale static motors. The regression rate determined through cylindrical motors is an approximation due to the lack of accurate real-time measurements. The regression rate in a cylindrical motor can be determined by measuring the fuel grain before and after the burn and noting the combustion properties throughout the burn. Such measurements are not instantaneous but are rather time- and space-averaged, leading to some errors [9,10]. Sacrificial regression rate measuring devices can be used for an instantaneous result. These sensors are cast within the fuel grain and make use of electrical resistance to determine their total height. As the fuel regresses, the combusting fuel erodes the sensor resulting in a change in resistance and an estimation of the fuel height. This technique has been tested [31] but was found to under-read the regression rate of the grain.

The regression rate data presented in this section include the averaged regression rate for an averaged oxidiser mass flux over the full burn time. Axial variation in regression rate is expected in hybrid combustion [48,49], but was not considered in the empirical regression rate data as the difference was either noted to be very small or not mentioned at all.

5.1. Slab motor testing

Slab motor test stands are a useful testing apparatus for both flow visualisation and regression rate analysis. Numerous institutions have constructed them with varying testing objectives for the development of hybrid technology. These objectives include either visualising the

combustion process through hot gas pyrolysis or full boundary layer combustion, or direct visual regression rate measurements. Hot gas pyrolysis allows the user to visualise the melt layer and any mechanical entrainment that may occur without any combustion. It can be used to define the amount of regression that is attributed to droplet entrainment [50].

Tables 1 and 2 include a summary of the existing slab motor test stands that have been developed primarily for paraffin wax and have yielded useful results to date. Each test stand has been used to capture specific data for the various grades of fuel. In some cases, significant regression rate data has been determined, particularly with the inclusion of additives. Table 1 addresses the design parameters, including the chamber geometry, fuel, oxidiser and any additives that were included in the fuel.

The French Aerospace Lab (ONERA) and Korea Aerospace University both investigated hot gas pyrolysis to assess the effect of liquid layer entrainment on regression [18,50–53]. By removing combustion, the effect of normal boundary layer combustion was eliminated, leaving only the effect of the unstable wave formation. In addition to this, Korea Aerospace University compared the pyrolysis result to full combustion results. Tokai University [41,42] and the Korea Aerospace University both investigated the effects of the fuel viscosity on liquid layer formation, and subsequent droplet entrainment. The viscosity was modified by mixing either EVA (ethylene vinyl acetate) or LDPE (low-density polyethylene) into paraffin wax respectively. Stanford University [54] used a slab motor to study droplet entrainment visually. For clear droplet visualisation, a space was included between the windows and the fuel grain. This introduced side burning but reduced the number of droplets that interacted with the window, distorting vision. The resultant slow motion videos clearly show droplet entrainment of the paraffin wax fuel into the oxidiser stream through the formation of waves in the liquid layer. The boundary layer formation and flame zone are clearly visible and useful in assisting with the understanding of the physical mechanisms at play during combustion. This test stand was not intended to measure fuel regression. Finally, SPLab [38] and Penn State University [25,30] both investigated the regression rate effects of fuels doped with metallised particles, specifically aluminium, magnesium and lithium hydride. The various pressure and oxidiser mass flux ranges can be seen in Table 2.

ONERA yielded an average regression rate result of 0.65 mm/s for pure paraffin wax under hot gas pyrolysis conditions. This is very low compared to commonly high results and is attributed to the lack of combustion regression [51]. The regression rate was measured through visualisation and an ultrasonic sensor, generating the instantaneous solid and liquid fuel thicknesses [50,59]. The ultrasonic measurements were reported, but due to the lack of combustion within the chamber, the gas temperature did not stabilise affecting the measured regression rate [52].

Both Tokai University and the Korea Aerospace University utilised slab motor test stands to develop an understanding of the effect of the fuel liquid layer viscosity of paraffin wax on regression rate, and if additives could be used to control this property [18,41,53]. In the case of the Korea Aerospace University, this was done with the intention of improving structural performance and was tested using both non-reactive hot gas pyrolysis and reactive boundary layer combustion.

Paraffin wax is known to be brittle, and there is a risk of fuel failure in the case of large scale vehicles [18,53]. The addition of LDPE was hypothesised to increase the tensile strength of paraffin wax. The regression rate of paraffin wax with 5% and 10% LDPE addition resulted in a progressive decrease in regression rate to that of pure paraffin wax. This decrease in regression rate still results in a higher overall regression rate to that of pure LDPE. Additionally, the LDPE doped paraffin wax samples showed an improvement in tensile and compressive strength of 24.8% and 34% respectively for the 5% doped samples, and 42.4% and 42.2% for the 10% doped samples [18,53]. The results obtained from Tokai University showed similarly that the addition of EVA in paraffin wax increases the melt layer viscosity, resulting in a lowered regression rate due to a reduction in droplet entrainment [41].

Pennsylvania State University and The Space Propulsion Lab (SPLab) from the Politecnico di Milano's Aerospace Engineering Department both used a double slab motor, which has a slab on the top and bottom of the combustion chamber, while still allowing for direct visualisation. The tests conducted by SPLab were extensive and included fuels such as gel wax, solid wax and HTPB, and additives at a wide range of oxidiser mass fluxes. The regression rate was determined based on the fuel volume and burn time [38] as follows:

$$\dot{r} = \frac{\Delta m}{\rho_f t_b A_b} \quad (2)$$

The SPLab tests were conducted on each fuel with varying percentages of nano-aluminium powders of 50 and 100 nm average particle size (named Alex50 and Alex 100 respectively) and MgH₂ powder (50–150 µm particle size) [38]. The results indicated a superior regression rate for the pure solid wax over the pure gel wax and HTPB fuels. At a mass flow rate of 350 kg/m²s, HTPB regression rate is measured to be about 0.6 mm/s, gel wax is 1.1 mm/s and solid wax is 2 mm/s [38]. Carbon black (0.2%) was added to the wax to increase the surface radiation heat transfer and limit the sub-surface radiation penetration, thus reducing the effect of sloughing [60]. The results indicate only a slight drop in the regression profile over the oxidiser mass flux range, while still also indicating the superiority of solid wax over the HTPB fuel. At 350 kg/m²s the blackened solid wax regression rate measures about 1.94 mm/s [38].

Table 1
Slab motor design parameters.

Institution	Fuel	Oxidiser	Additives	Wall interaction	Slab dimensions (mm)	Windows
ONERA [50–52]	Paraffin wax (C31H64)	None (pyrolysis)	None	No	40 × 24.82 × 350	Yes
TU [41,42]	Paraffin wax (C51H104)	GOX	EVA (C5H7O2(CH3))	Yes	10 × 10 × 100	Yes
KAU [18,53]	HDPE, LDPE, Paraffin wax (C28H58)	GOX	LDPE	Yes	Width of 50 mm	Yes
SU [54]	Paraffin wax (C32H66), HTPB and HDPE	GOX	Carbon Black	No	25 × 9.5 × 127	Yes
SPLab [38]	HTPB, Gel wax (C12H26), Solid wax (C24H50)	GOX	Nano-aluminium (Alex50) and (Alex 100), MgH ₂ powder (50–150 µm particle size) and LiH	Not specified	10 × 4 × 50	Yes
PSU [25,30]	HTPB	GOX	UFAL	Yes	76.2 × 42.8–44.5 × 584.4	None
DLR [55–58]	Sasol Wax 6003, 6805, 0907, 1276	GOX	Carbon Black, SA, Nanoclay, Common Polymer	Not Specified	90 × 14 × 180	Yes

*ONERA – Le centre français de recherche aérospatiale * SU – Stanford University * DLR – Institute of Space Propulsion * TU – Tokai University * KAU – Korea Aerospace University * SPLab – Politecnico di Milano * PSU – Pennsylvania State University.

Table 2
Slab motor operating conditions.

Institution	Operating pressures (bar)	Oxidiser mass flux range (kg/m ² s)	Regression error interval	Testing objective
ONERA [50–52]	10–30	50–300	Not stated	Liquid layer visualisation, and effect on regression rate through hot gas pyrolysis. Regression rate measured with window visualisation with high-speed cameras and ultrasonic sensor, generating the instantaneous solid and liquid fuel thicknesses.
TU [41,42]	Atmospheric	10–30	Not stated	Control the regression rates of fuels by changing their viscosities with the use of additives. Regression rate determined through window visualisation with high-speed cameras
KAU [18,53]	1.1–2.6	4–15	Not Stated	Manipulate the liquid layer viscosity of paraffin wax fuels. Tested with both hot gas Pyrolysis and Full boundary layer combustion.
SU [54]	Up to 17.2	Up to 40	Regression not measured	Visualise the droplet entrainment in liquefying fuels
SPLab [38]	1.5	100–350	Not stated	Double slab (top and bottom) configuration motor for testing the regression rates of various fuels with additives
PSU [25,30]	23–46	175–225	±5%	Double slab (top and bottom) configuration motor for testing the regression rates of different fuels with performance additives. Thermocouples were embedded into the fuel during some of the tests to measure surface temperature during combustion
DLR [55–58]	Atmospheric	6–8	±10%	Determine a relationship between liquid layer viscosity and regression rate.

*ONERA – Le centre français de recherche aérospatiale * SU – Stanford University * DLR – Institute of Space Propulsion * TU – Tokai University * KAU – Korea Aerospace University * SPLab – Politecnico di Milano * PSU – Pennsylvania State University.

With the addition of 5% Alex100 aluminium particles to solid and gel wax, the regression rates at 350 kg/m²s increased to 2.2 mm/s and 1.9 mm/s respectively. There was a substantial increase in regression performance of the gel wax (71%), and only a slight increase in performance for the solid wax (10%) when compared to their pure forms [38]. Regression rates with the addition of 5% magnesium hydride (MgH₂) into HTPB, gel wax and solid wax at 350 kg/m²s were measured to be about 1.28 mm/s, 1.64 mm/s and 2.5 mm/s respectively. The performance increase for the solid wax and gel wax (relative to the pure fuels at 350 kg/m²s) was 38% and 25% respectively. These results indicate that the performance increase with the addition of MgH₂ has a lower dependency on the binder material [38,61,62].

Pennsylvania State University used their slab motor for analysing the regression rate of HTPB with the addition of ultrafine aluminium

particles (UFAL). These particles have an average size of 0.05–0.10 μm. This data was later correlated to other cylindrical motor tests and showed very similar results [25,30]. Results from these slab motor tests indicate a large regression rate advantage for fuels with certain additives, which is a similar conclusion to that drawn by SPLab. The addition of 20% (by weight) UFAL to HTPB increased the fuel mass flux by 70% [25]. The regression rate data obtained from the Institute of Space Propulsion revealed a slight increase in regression rate with the addition of 2% nanoclay, but a significant decrease in regression rate with 5 and 10% of a common polymer. The regression rate data was measured on a short burn slab motor, and therefore the regression rates are not comparable to the results presented in Fig. 3 due to the large ignition transients and should only be used for comparative tests on the same test apparatus [55–58]. The regression rate data obtained from some of these slab motor tests have been plotted on a common set of axes for comparison, as shown in Fig. 3.

5.2. Laboratory-scale motor testing

Laboratory (Lab)-scale motor testing is one of the most commonly used methods of generating regression rate data. The advantage over a slab motor is the use of a more realistic grain geometry. Regression rate is often determined by time-averaging based on the grain dimensions before and after the burn time, as well as the averaged combustion properties. Numerous institutions have conducted lab-scale motor testing with varying objectives behind their motor designs and testing methodologies. These design attributes are given in Tables 3 and 4. The data within these tables were extracted from a number of sources, and is either directly specified as a test condition, or indicated as a design range of the equipment. A shared objective of most of the tests conducted was the determination of the performance characteristics of a propellant combination, and the effect various additives have on the fuel regression rate.

Table 3 includes the different geometrical properties of the lab-scale motors designed. Some of the tests conducted were either on the same or modified versions of existing lab-scale motors. Table 4 lists the operating conditions and testing objectives. Before the inception of paraffin wax as a hybrid fuel, a common hybrid fuel was HTPB [37]. Significant research went into the regression rate improvement of this fuel, in particular with the inclusion of energetic additives. Since the advantages of liquefying fuels were discovered, researchers began to investigate the improvement of paraffin wax over a doped HTPB fuel [23,26,30,63]. Finally, to further the regression rate improvement of liquefying fuels, energetic additives were then added to paraffin wax. This gradual development of fuel regression rate improvement has advanced hybrid rocket fuels substantially.

The regression rate data extracted from the publications referenced in Tables 3 and 4 is presented graphically in Figs. 4 and 5. Fig. 4 includes the pure paraffin and blackened paraffin results from each institution. A trend line has been used to represent the point results from Doran et al. [64]. This work did not present a regression curve due to insufficient and inconclusive tests. The data was included regardless, as it was the only pure paraffin wax data graphically represented that makes use of N₂O as the oxidiser over a range of oxidiser mass fluxes. The University of Brazil [41] has conducted tests on pure paraffin wax with nitrous oxide, but this data is tabulated, and the oxidiser mass flux is uncertain due to doubt in the flux measurement technique. The remaining tests make use of gaseous oxygen (GOX).

The regression rate data for most of the tests result in a similar regression profile except for the two distinct outliers from Larson et al. [65]. These two datasets, although obtained from the same LGCP motor as Evans et al. [30], had a critical adjustment to the grain dimensions while keeping the motor dimensions the same. The grain only made up about one-quarter of the chamber length, affecting the flow and regression properties. The short burn times and small diameter design of this motor made its design purpose for comparative tests only [65]. Additionally, there was a difference between the Pennsylvania State

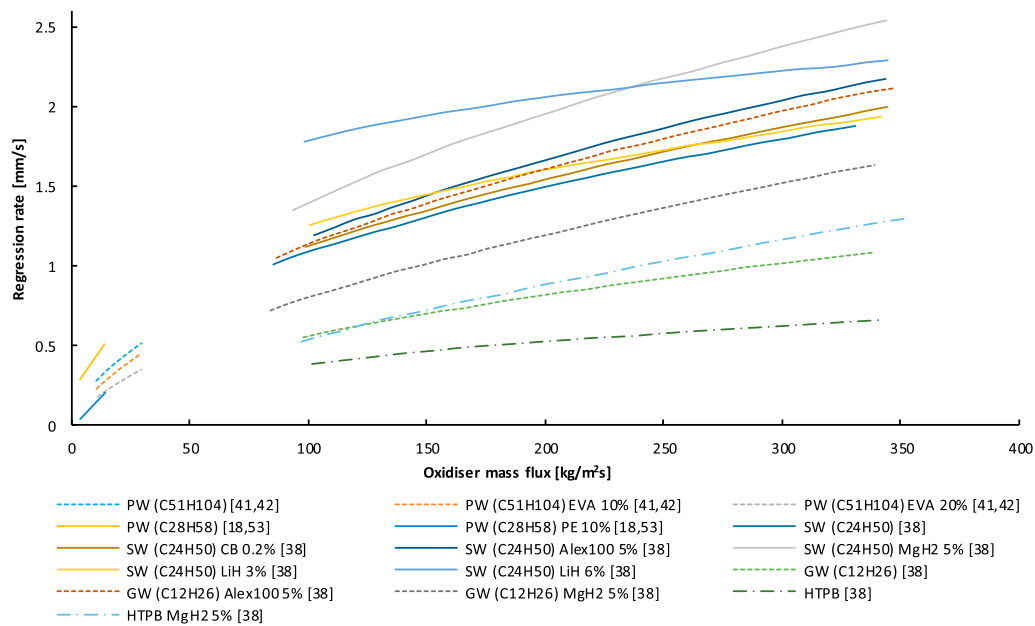


Fig. 3. 2D regression rate results for various wax samples with additives.

Table 3
Lab-scale motors design parameters.

Institution	Fuel	Oxidiser	Grain OD (mm)	Grain ID (mm)	Grain Length (mm)	Chamber Length (mm)	Regression error interval
SU [11,66]	Paraffin Wax (C32H66), Plexiglass, HTPB	GOX	60	12.7–30.48	177.8	177.8–482.6	Not stated
NASA AR [67,68]	Paraffin Wax (C32H66), Plexiglass and HTPB	GOX	195	76.2–152.4	838.2–1117.6	1200	±3–8%
SU [69]	HTPB, PMMA, HDPE, Sorbitol	N ₂ O	50.8	19.05	127, 152.4 and 228.6	152.4, 177.8, 254	0.044
SU [64]	Paraffin Wax (C32H66), HDPE, PMMA, HTPB	N ₂ O	50.8	12.7–19.05	101.6–228.6	up to 254	Not stated
UB [70,71]	Macrocrystalline Paraffin wax	N ₂ O	71	23	220	–	Not stated
UB [72]	Macrocrystalline Paraffin wax	N ₂ O	71	23	220	–	Poor results
IIT [73]	MW-704 Paraffin Wax, PMMA	GOX	21	190	310	–	±8%
KAU [53]	Paraffin Wax (C28H58), LDPE, HDPE	GOX	70	20	200	–	Not stated
PSU [23,26,30,63]	Paraffin Wax (C32H66), HTPB	GOX	38	8.86	410	–	±3%
PSU [30,63] XTC	Paraffin Wax (C32H66), HTPB	GOX	114	64	457	–	±3%
PSU [65] LGCP	Paraffin Wax (C32H66), Paraffin Wax (Aerospace Corporation)	GOX	38	8.89–12.97	101.6–139.7	–	±3–7%
UN [74]	HTPB, Paraffin Wax (C24H50)	GOX/ N ₂ O	69.2	19.1–28.8	220–240	350	Only defined for failed tests
IITM [75,76]	30% Micro-Crystalline wax, 70% Paraffin Wax	GOX	42	9	134	134	±2.4–4%
IITM [75]	30% Micro-Crystalline wax, 70% Paraffin Wax	GOX	100	20	300	300	±1.7–2.5%

*SU – Stanford University * NASA AR – NASA Ames Research Centre * PSU – Pennsylvania State University * UB – University of Brazil * IIT – Israel Institute of Technology * KAU – Korea Aerospace University * UN – Università di Napoli * IITM – Indian Institute of Technology Madras.

University and Aerospace Corporation paraffin wax compositions. The paraffin wax/N₂O experimental data plotted as a trend line from the set test points is noted to be much higher than expected for a nitrous oxide propellant when compared to the GOX propellants [64]. There is uncertainty in the regression rate data measurements from this set of tests [64]. The remaining regression profiles correlate closely to each other and do not seem to be affected by the carbon chain length of the known waxes used.

The data presented by Karabeyoglu et al. [11,66], investigated transient combustion within hybrid rocket motors. A choked sonic orifice placed in the feed line was used to control the mass flow rate. The regression rate data were obtained for very low oxidiser mass fluxes but remains useful in indicating the regression rate advantage of paraffin wax over a standard HTPB/GOX system. The NASA Ames Research Centre

established a scaled-up test facility for further regression rate testing at higher oxidiser mass fluxes [67,68]. Their results indicate that scale has little effect on the regression rate of paraffin wax, thus small-scale motor data can be applied to larger scale motors. Also, the oxidiser mass fluxes and chamber pressures observed in these scaled-up tests are more realistic when compared to those expected in a flight motor. Additional data obtained from these scale-up tests indicate that grain length and chamber pressure do not affect the regression rate of the fuels [68].

Rajiv and Ramakrishna from the Indian Institute of Technology Madras conducted lab-scale motor testing with the intention comparing combustion chamber pressure and weight change regression rate measuring techniques [75,76]. Their tests were conducted on two motors of 134 mm and 300 mm grain length. The regression rate measurement techniques correlated well to existing data. However, there is a

Table 4
Lab-scale motors operating conditions.

Institution	Additives	Operating pressures (bar)	Oxidiser Mass flux range (kg/m ² s)	Objective
SU [11,66]	None	3.8–13.8	5–150	Lab scale motor was designed to fulfil the experimental requirements of determining the transient combustion properties, as well as the regression rates of liquefying hybrid fuels
NASA AR [67,68]	None	10–68	74.3–368.7	Designed as a scaled up version of the above lab scale motor to determine the effect pressure, oxidiser mass flux, and fuel grain length have on the regression rate
SU [69]	None	20–45	25–270	Lab scale motor designed to characterise the regression rate of traditional and novel hybrid fuels using nitrous oxide as the oxidiser
SU [64]	5 µm aluminium	27.5–48	25–300	Utilising the same lab scale motor as above, these tests were conducted to characterise the effect of aluminium particles on the regression rate of some hybrid fuels.
UB [70,71]	Carbon black	12–20	200–700	This lab scale motor was developed with the intention of investigating and comparing the ignition of solid fuels, regression rates, and combustion performance of paraffin wax with nitrous oxide and HTPB and gaseous oxygen
UB [72]	Carbon black	12 - 20.	96–585	The further testing conducted on the above lab scale motor was conducted with the intention of presenting a mathematical correlation for the combustion velocity as a function of the oxidiser mass flux.
IIT [73]	None	12 - 18.	5–100	Experimental testing conducted for the correlation of test data to a theoretical model for liquefying fuels
KAU [53]	LDPE	Unknown	33–475	Testing conducted to determine the performance of paraffin wax doped with varying percentages of LDPE with the intention of increasing the structural properties of paraffin wax.
PSU [23,26,30,63] LGCP	Carbon black in paraffin wax, aluminium in HTPB	23–46	140–850	Compare the differences between aluminium loaded HTPB and paraffin wax as a means to improve regression rate in hybrid rockets
PSU [30,63] XTC	Silberline nano-sized aluminium in paraffin wax	21.7–42.4	163–320	Determine the regression rate of aluminised paraffin wax fuel grains.
PSU [65] LGCP	Lithium aluminium hydride (LiAlH ₄), cyclotrimethylenetrinitramine (RDX) (C ₃ H ₆ N ₆ O ₆), and multi-walled carbon nanotubes (MWNT).	13.8–24.1	49–115	The addition of LiAlH ₄ and RDX for regression rate and performance enhancement. Addition of MWNT for structural enhancement
UN [74]	SEBS in Paraffin Wax, aluminium powder in HTPB	4–18	30–130	Compare the differences between aluminium loaded HTPB and paraffin wax with SEBS as a means to improve regression rate in hybrid rockets
IITM [75,76]	None	9	- 160	Comparing regression rate measuring techniques. Weight measurement vs. pressure measurement
IITM [75]	None	9	- 350	Measure the regression rate

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noticeable difference as a result of scale. The fuel used in their testing was a mixture of 30% microcrystalline and 70% paraffin wax. The carbon chain length was not noted in their publications.

For the purposes of this review, further regression rate data was obtained with the inclusion of a variety of additives within a paraffin wax binder and other novel hybrid fuels. The various regression data from Tables 3 and 4 are given in Fig. 5 to allow for the graphical comparison of the effects of additives. Stanford University designed a new lab-scale motor for the regression rate testing of HTPB, polymethyl methacrylate (PMMA), high-density polyethylene (HDPE) and sorbitol [69]. This apparatus was later used for the regression rate testing of aluminised grains, as well as paraffin wax [64]. The system was designed to be modular in an attempt to fit different grain diameters and lengths, accommodating a variety of fuels. Tests were conducted with nitrous oxide as the oxidiser due to its safety advantages over GOX. Trend lines for the 5, 10, and 20% aluminised paraffin wax fuels grains are given in Fig. 5 [64] and compared against the most conclusive pure paraffin wax regression sample from Karabeyoglu et al. [67,68]. The scattered data does not show a conclusive increase in regression rate with each

percentage of aluminium. Doran et al. [64] attribute this to the short burn times and few results. A similar conclusion was drawn for the pure paraffin results represented in Fig. 4.

Other energetic additives such as LiAlH₄ were tested by Larson et al. [65]. Results of a 5 and 10% addition of this hydride are plotted in Fig. 5. There is a 7–10% increase in the regression rate of this fuel over that of the pure paraffin tests conducted. Larson et al. [65] attribute a portion of this increased regression rate to the lower fuel density when compared to pure paraffin wax. However, there is still an increase in fuel mass burn rates in fuels with additional LiAlH₄. Evans et al. [30] demonstrated a 30% increase in regression rate with the inclusion of 13% Silberline aluminium particles. It was hypothesised that the aluminium flakes are encapsulated with molten wax when they leave the fuel surface and burn rapidly near the fuel surface promoting the ignition of the aluminium particles.

Kim et al. [53] conducted tests to increase the structural performance of the paraffin wax with the addition of LDPE. Fig. 5 indicates the resultant regression rate data, showing a reduction of the average regression rate at increasing percentages on LDPE. The addition of LDPE

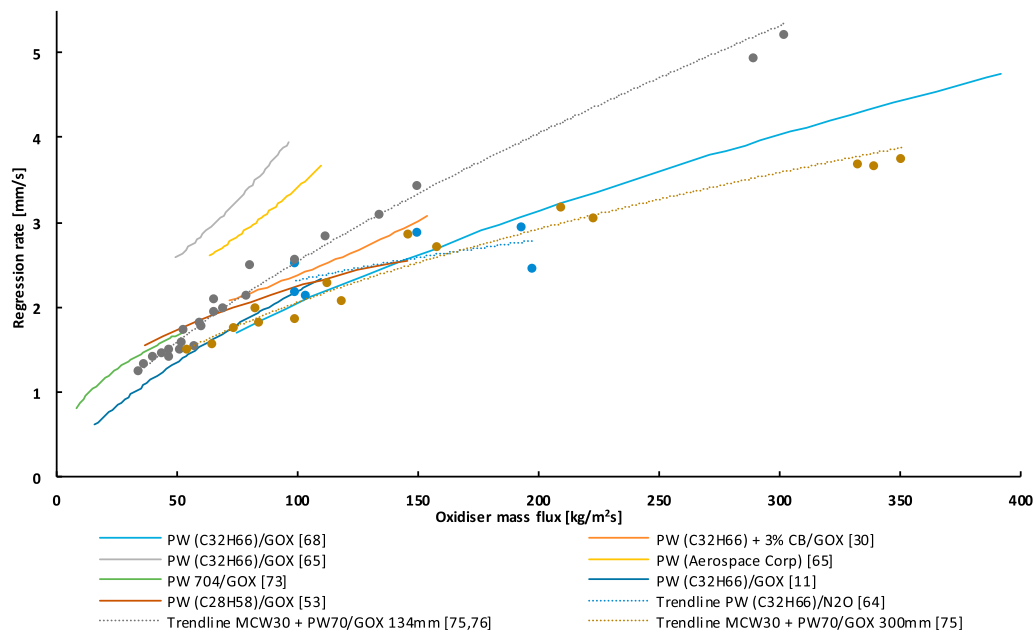


Fig. 4. Regression rate results for pure paraffin wax samples.

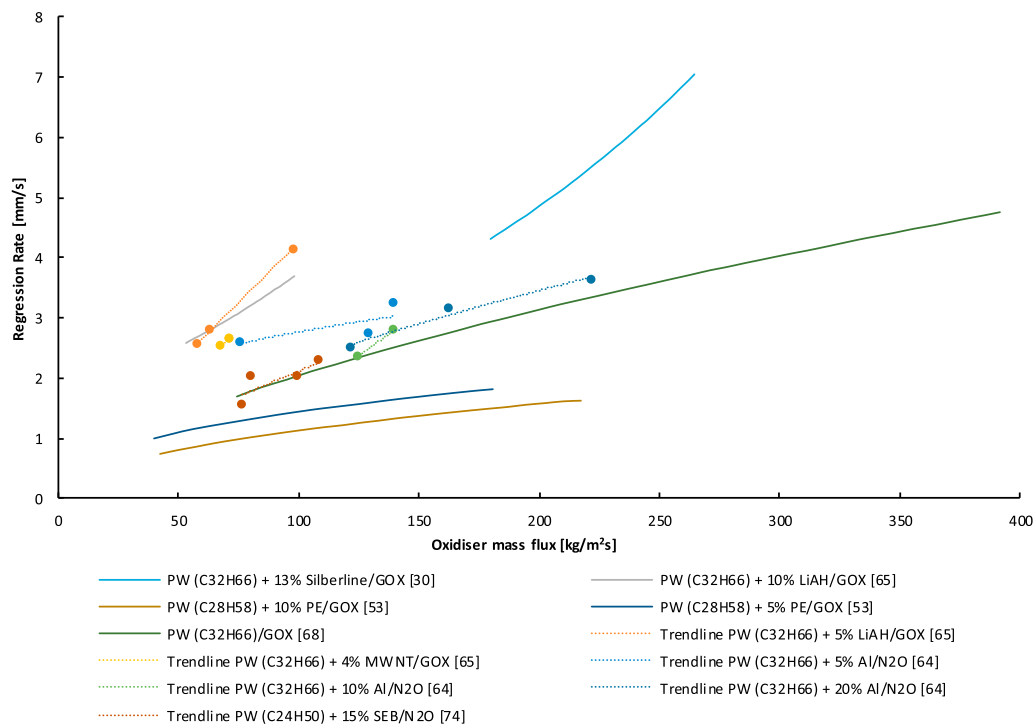


Fig. 5. Lab-scale regression rate results for various wax samples with additives.

increased the liquid layer viscosity, reducing the regression due to droplet entrainment. Scaramuzzino et al. [74,77] added an energetic additive referred to as SEBS, which is a thermoplastic polymer. These test results indicated a higher regression rate when compared to aluminised HTPB, however many of the results were uncharacteristically high due to structural failure of the grain, resulting in fragment loss. A new casting method was then considered which reduced the fragment loss occurrence, but the wax still did not meet structural requirements.

Other performance testing with the inclusion of aluminium particles includes the work done by Sun et al. [78] and Li et al. [79] where the regression rate of aluminium and aluminium and magnesium doped

HTPB fuels grains were tested. The results show a noticeable increase in the regression rate of the fuel with up to 35% Al, and 10% Mg over that of pure HTPB.

The results presented indicate similar regression trends, with most metalized additives improving the regression rate, polymer additives reducing the regression rate, as well as a general increase in regression rate with an increasing oxidiser mass flux. There is unfortunately, very little consistency across testing platforms, due to the differences in measurement methods, averaging techniques and testing procedures. Rajeshwar and Gany [89] investigated the effects of similarity and scaling in hybrid testing and concluded that for a test motor to accurately

simulate a full-scale motor, geometric similarity, oxidiser fuel combination, and scaling of the oxidiser mass flux to the port diameter is required. Further, recent testing conducted by Cai et al. demonstrated the significant effect the length-to-diameter ratio has on the regression rate and combustion efficiency [90] of a hybrid motor. It is advised that a set of testing standards be determined for regression rate testing of hybrid rocket motors so that results can be reasonably compared.

6. Structural assessment of fuel grains

Both HRMs and SRMs make use of a solid fuel the combustion chamber. This solid fuel mass is geometrically designed to meet the mission thrust requirements. There are numerous publications available defining the structural assessment of a SRM due to the high-risk nature of a failure [2,80–83]. The development of a crack in a solid propellant can result in motor over pressurisation or vehicle loss [84]. There is, however, very little information on the structural performance of hybrid fuels grains. This has not been a primary concern in the development of hybrid technology. Hybrid fuel cracks do not carry as much risk of explosive failure as those in solid fuels. The separation of the fuel and oxidiser limits the fuel reaction to surfaces directly exposed to the oxidiser stream. Further, liquefying fuels may offer a crack healing phenomenon, where the liquid fuel is allowed to flow into any void created as a result of structural failure. The primary failure concern in hybrid fuels is the loss of fuel through the nozzle, which is then at risk of damage or of nozzle blockage resulting in a partial mission altitude [2]. Multiport fuel grain designs, commonly used with classical hybrid fuels are at high risk of failure due to sliver fracture of thin-walled ribs. High regression fuels such as paraffin wax do not require multiple ports and can offer some structural rigidity with a single port geometry. However, the brittle nature and low material strength of paraffin wax create uncertainty in its ability to withstand launch loads in its pure form.

Paraffin wax is known to be a soft, weak material. Its low strain failures make its use in high load launch situations concern for structural failure [34,85]. Karabeyoglu [47] and Cantwell et al. [16] have commented on the negative impact that metal additives will have on the structural properties of paraffin wax; however little work has been conducted to verify this.

6.1. Material testing

Characterising the structural response of propellants is necessary to determine the feasibility of a fuel for the launch conditions, and numerous sources have conducted tensile tests on paraffin wax fuels, with additives. A full characterization is required to determine the temperature and strain rate dependency, but most tests to date are less comprehensive than this. The lab scale tests conducted by Scaramuzzino et al. [74,77] documented many fragment losses due to structural failure which resulted in nozzle damage. They attributed this failure to the poor mechanical properties of the SEBS doped paraffin wax grain as a consequence of the low-temperature casting method. Additives such as LDPE have been added to paraffin wax with the intention of improving the structural performance. The effect of this material is to lower the regression rate, as discussed. DeSain et al. [86] conducted structural tests on paraffin wax with 0–4% LDPE addition following the ASTM 638-03 international standard. The results indicated that LDPE increases the stiffness of the material, and increases the ultimate tensile strength (UTS) to a similar range of HTPB. Unfortunately, the stiffness is higher than HTPB, resulting in a fragile grain. They determined the optimal LDPE addition to be 2%, which increases the UTS of paraffin wax to an average of 2.5 MPa at a strain rate of 5.1 mm/min [86]. This is an improvement of 150% over the pure paraffin wax average UTS of 0.94 MPa. Kim et al. [53] conducted tests of a similar nature but followed the procedure stipulated in ASTM D1320-73. This standard has a vastly different testing geometry and has been withdrawn with no replacement. LDPE percentages of 5 and 10% were tested in compression and tension. Similar to

DeSain et al. [86] the UTS increased with increasing proportion of LDPE. The average UTS at 0, 5 and 10% LDPE is 1.57, 1.96, and 2.23 MPa respectively at a rate of 5 mm/min.

Kobald et al. [57] conducted extensive mechanical testing on various paraffin wax formulations (Sasol, 6003, 6805, 0907, 1276) both macro and microcrystalline, with the additives such as nanoclay, stearic acid (SA), and what is referred to as a common polymer. The results presented offer the tensile properties of Sasol 6805 with the various additives, normalised to Sasol 6805 + 10% SA. A 2% addition of nanoclay doubles the elongation of the wax but does not significantly improve the tensile strength. It does however slightly increase the regression rate. The addition of 5 and 10% polymer increased the strength by two and three times respectively and the elongation by three and five times respectively. The regression rate of the paraffin wax doped with the polymer had a reduced regression rate. Finally, the addition of a combination of 5% polymer and 2% nanoclay gave an increase similar to that of the 10% polymer [57]. A similar evaluation of the effects of organically modified montmorillonite (OMMT) on the structural properties of paraffin wax was conducted by Wang et al. [85] where percentages between 0 and 5% were added to paraffin wax. The structural properties of pure paraffin wax were identified as a ductile failure at low strain rates. The addition of 2% OMMT increased the elongation by 450% [85].

Other additives such as EVA were structurally tested. The addition of 0–20% EVA was characterised by Maruyama et al. [42] who followed the JANAF standards for tensile tests. These tests yielded an increasing UTS with an increasing percentage of EVA addition. The addition of 20% EVA increased the UTS by 1.6 times and the strain by 2.3 times [42]. The effects of the structural properties of paraffin wax grains with aluminium particles were investigated by Veale et al. [87] and Ryu et al. [88]. Veale et al. noted a slight increase in the UTS of Sasol 0907 with a 15% drop in elongation with the 40% aluminised grain. Ryu et al. considered the effect of vary percentages of nano- (100 nm) and micro- (8 µm) sized aluminium particles at varying weight percentages. It was noted that the grains doped with nano-AL displayed a noticeable improvement in structural properties, while the micro-AL doped grains showed less of an improvement, although the tensile strength was still improved.

Table 5
Lab-scale motors design parameters.

Sample (+Wt% additive)	Pull rate (mm/min)	Average UTS (MPa)	Experimental standard followed
PW [86]	5.1	0.9395	ASTM D638-03
PW	508	0.8915	ASTM D638-03
PW + 2% LDPE	5.1	2.5	ASTM D638-03
PW + 4% LDPE	5.1	1.81	ASTM D638-03
PW + 5% PE	5	1.96	ASTM D1320-73
[18,53]			
PW + 10% PE	5	2.23	ASTM D1320-73
PW [85]	10	0.77	ASTM D638-95
PW + 0.5% OMMT	10	0.91	ASTM D638-95
PW + 1% OMMT	10	1	ASTM D638-95
PW + 2% OMMT	10	1.06	ASTM D638-95
PW + 3% OMMT	10	1.15	ASTM D638-95
PW + 4% OMMT	10	1.13	ASTM D638-95
PW + 5% OMMT	10	1.22	ASTM D638-95
PW [88]	5	2.118	ASTM D638-95
PW + 10% Nano Al	5	2.759	ASTM D638-95
PW + 20% Nano Al	5	3.068	ASTM D638-95
PW + 30% Nano Al	5	2.764	ASTM D638-95
PW + 5% Micro Al	5	1.768	ASTM D638-95
PW + 10% Micro Al	5	2.014	ASTM D638-95
PW + 15% Micro Al	5	2.601	ASTM D638-95
PW [42]	1	0.8	JANAF
PW + 10% EVA	1	1.1	JANAF
PW + 20% EVA	1	1.42	JANAF
PW [87]	1	1.73	JANAF
PW + 40% AL [87]	1	1.86	JANAF

Table 5 lists the average UTS obtained from these sources. It can be noted that there are no sources that have conducted a conclusive characterization of the material. Most cases include room temperature tensile and occasionally compressive tests. In this case, each of the sources made use of different standards for testing, none of which are specifically suited to this material.

The results available are not sufficient to accurately represent the loading conditions that a hybrid rocket fuel grain would encounter during flight. It is advised that once a suitable additive is found to meet the performance requirements, comprehensive structural testing is conducted to characterise the nature of the propellant fully. These tests should follow similar procedures used in solid rocket motor analysis, and include properties such as strain rate dependence, temperature dependence, and creep to name a few.

7. Conclusion

Hybrid fuels, particularly liquefying fuels, have shown potential as an applicable form of chemical propulsion. Despite the advances in this technology, no detailed review of the existing performance and structural test results or methods are available for paraffin wax in particular. In this work, an extensive review of paraffin wax hybrid fuel structural properties versus its regression rate has been presented. This was done with the intention of creating a link between existing structural additives, and performance measures. The density impulse in conjunction with the regression rate as a performance measure is a good indication of the expected performance of a propellant combination in volume-limited designs.

The regression rate data presented indicates a strong link between regression rate improvements in paraffin wax with the addition of metalized additives. Unfortunately, very little testing has been conducted in the structural performance of the grains with these additives. The structural tests that have been conducted have been done under varying conditions, with numerous inconsistencies between sources. In general, there seems to be an improvement in the UTS of the material with small percentages of aluminium, while some report an improvement in elongation, and other report a reduction in elongation. Paraffin wax with a small percentage of a polymer additive, such as EVA or LDPE was investigated with the objective of improving on the materials naturally brittle nature. Small percentages of both EVA and LDPE showed an increase in the UTS and the elongation of paraffin wax. Performance testing of these additives resulted in a reduction in the regression rate. This was due to the increased liquid viscosity, reducing the number of droplets that are entrained into the oxidiser stream. This has been verified in 2D slab motor tests. The structural tests performed have only been conducted at room temperature and at slow speeds, limiting the use of the results. High strain rate tests and temperature dependence of the material should be considered for in-flight loading conditions, while creep and relaxations testing should be considered for storage related structural analysis.

The data presented in this paper indicates gap in the development and use of paraffin wax hybrid fuels for launch applications. It is important to consider both the propellant performance, and its ability to withstand the loads induced during launch. A fuel structural analysis requires comprehensive material characteristics before the structural feasibility can be determined. When considering a new additive, the fuel formulation should be tested with considerations to strain rate dependence and temperature effects. Additionally, the performance of any propellant combination needs to be determined. There are numerous inconsistencies between the test procedure presented above, including chamber size, fuel dimensions, burn times, measurement techniques, averaging methods, and finally the number of tests conducted.

The next step in the development of paraffin wax based hybrid fuels and their feasibility in propulsion is to develop a standard procedure for testing which considers which variables need to be reported, how they should be measured, and finally how they should be reported. The

ballistic coefficients, a and n , in hybrid propulsion can often only be determined empirically, and then used to model flight size motors. These flight size motors also need to carry a structurally sound solid fuel grain, which shows no risk of in-flight cracking or failure. This is a similar process to that used in solid rocket propellants, which should be applied to enhance the future development of hybrid fuels.

Acknowledgements

The Authors would like to acknowledge the financial support afforded through the National Research Foundation - *Thuthuka* Grant 84460.

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